STRUCTURE ABSTRACT

Background : Kerala has emerged as the epicenter of coronary artery disease (CAD) in South Asia, with prevalence rates substantially exceeding the national average. The syndemic burden integrates genomic predispositions --- including ACE insertion/deletion polymorphisms, PCSK9 gain-of-function variants, MTHFR C677T, HMGCoA reductase SNPs, and diabetes/obesity susceptibility loci --- with epigenetic dysregulation driven by early-life dietary insufficiency, environmental oxidative stress, and socioeconomic adversity. Crucially, CAD manifests a decade earlier in Keralites than in Western cohorts, disproportionately afflicting the 20-40-year productive age group.

Objectives: GREENAGE-KL aims to (1) establish early molecular CAD risk detection using serum, Fatty Acid Binding Protein (FABP) quantifies by near-infrared spectroscopy as a novel pre-clinical biomarker; (2) evaluate sulforaphane-rich microgreens (broccoli, red radish, fenugreek; 20-50 g/day) as epigenetic modulators targeting DNA methylation, histone acetylation, and NRF2/KEAP1-axis oxidative stress reduction; and(3) integrate a dual-layer Shannon Codon Entropy (J') framework --- correlating 64-codon degeneracy scores with biological resilience --- into a precision nutrition AI platform.

Methods: A 12-month prospective marched-cohort study enrolling 500 participants (250 with one or more CAD risk factors; 250 age-sex-matched controls), aged 20-40 years, across 14 Kerala districts. Layer 1 employs AHA PREVENT cardiovascular risk scores, cardiac echocardiography (LVEF, GLS, diastolic parameters), CardioSense HRV analysis (SDNN, RMSSD, Shannon entropy, triangular index, frequency and time domain metrics --- FDA-approved), and biochemical oxidative stress profiling (malondialdehyde, superoxide dismutase). Layer 2 deploy a synthetic multi-omics Shannon codon degeneracy index (J' = 0-100%) derived from 20 amino acid codon families across 100 synthetic Kerala representative genomic data points from all 14 districts.

Expected Outcomes: Demonstration that sulforaphane-driven epigenetic reprogramming, quantified by rising Shannon J' indices, can retard subclinical CAD progression by 5-10 years in genetically susceptible young Keralites. Delivery of a fully functional HRIDAYALAYA Genomic Precision Platform integrating HER, FABP spectroscopy, HRV, multi-omics, and AI-driven nutritional prescription.

Keywords: *FABP spectroscopy · sulforaphane · epigenetics · Shannon codon entropy · CAD prevention · Kerala · microgreens · NRF2 · HRV · precision cardiology*

1. BACKGROUND & SCIENTIFIC RATIONALE

1.1 The Kerala CAD Crisis

Kerala, despite its superior health indicators (literacy, life expectancy, IMR), paradoxically leads India in cardiovascular morality. CAD prevalence in Kerala exceeds 14-15% in urban populations and 12% in rural cohorts --- nearly double the national average. The INTERHEART South Asia substudy confirmed that South Asians develop CAD 5-10 years earlier than Western counterparts, and Kerala-specific data indicate early-onset CAD (age 20-45) constitutes 25-30% of all acute MI presentation at tertiary centres.

1.2 Genomic Architecture Favouring CAD in Kerala

Four genomic axes confer excess CAD risk in the Kerala population:

- Renin-Angiotensin System: ACE gene insertion/deletion (I/D) polymorphism – DD homozygotes exhibit 2-fold elevated cardiovascular risk with heightened aldosterone-driven hypertension susceptibility.
- Cholesterol Metabolism: PCSK9 gain-of-function variants and HMGCoA reductase SNPs create pharmacogenomic barriers to statin efficacy in Kerala cohorts.
- Methylation Deficit: MTHFR C677T TT genotype (prevalent ~18% in Kerala) drives hyperhomocysteinaemia, endothelial dysfunction, and impaired folate-mediated epigenetic methylation.
- Metabolic Vulnerability: TCF7L2, PPARG, FTO, and MC4R variants aggregating in Kerala families predispose to insulin resistance, visceral adiposity, and diabetogenic dyslipidaemia.

1.3 Epigenetic Dysregulation as the Amplifier

Genomic risk is necessary but not sufficient. Epigenetic modifications --- DNA methylation at CpG islands of promoters (NRF2, CDKN2A, RASAL1), histone deacetylation (HDAC-mediated silencing of cardioprotective genes), and microRNA-33/126/155-driven post-transcriptional regulation --- convert latent genomic risk into active CAD pathobiology. Poor dietary methyl donor (folate, betaine, choline), high refined carbohydrate consumption, psychosocial stress, and environmental pollution (particularly in costal and industrial Kerala districts) synergise to accelerate epigenetic ageing beyond chronological age.

1.4 Fatty Acid Binding Protein (FABP) as Early Molecular Sentinel

Heart-type FABP (H-FABP/FABP3) is a 15 kDa cytoplasmic protein released within 30-60 minutes of cardiomyocyte injury, preceding troponin by 3-4 hours in acute settings. More critically, circulating FABP levels reflect subclinical myocardial lipid metabolic stress BEFORE ischemia-driven necrosis. Near-infrared (NIR) spectroscopy enables label-free, rapid quantification of FABP with sensitivity in the sub-nanomolar range, avoiding immunoassay latency. This study incorporates FABP-NIR spectroscopy as the primary early molecular biomarker --- a paradigm-shifting addition to CAD risk stratification.

1.5 Sulforaphane: Epigenetic Precision Nutrient

Sulforaphane (SFN), the isothiocyanate derivative of glucoraphanin hydrolysed by myrosinase in cruciferous microgreens, is the most potent natural activator of the NRF2 transcription factor. NRF2 nuclear translocation upregulates ARE-driven antioxidant enzymes (HO-1, NQO1, GCLC, TXNRD1), reduces reactive oxygen species (ROS) and malondialedehyde (MDA), and modulates DNMT/HDAC activity to restore epigenetic balance. Broccoli microgreens contain 10–100× higher glucoraphanin concentrations than mature heads. This study employs standardised microgreen doses (20-50 g/day) from HRIDAYALAYA's certified cultivation facility --- ensuring compositional consistency of sulforaphane yield across the trial.

1.6 Shannon Codon Entropy (J') --- The Biological Chaos-to-order Framework

Shannon entropy $H = -\sum p_i \log_2 p_i$, when applied to codon degeneracy across the 64-codon genetic code, generates a Pielou's evenness index (J'). The 20 amino acids (2 with single codon, 3 basic amino acids --- Lys, Arg, His, 15 neutral amino acids) exhibit differential degeneracy: maximal codon diversity ($J' \rightarrow 1.0$) represents biological resilience and redundancy buffer against point mutations, while J' collapse ($\rightarrow 0$) predicts vulnerability to oxidative and metabolic stress. The GREENAGE-KL framework hypothesizes that nutritional-epigenetic interventions (sulforaphane, microgreens) measurably shift J' trajectories upward by modifying codon-usage bias through DNMT-medicated methylation patterns --- a testable, falsifiable mechanistic hypothesis.

AIM & OBJECTIVES

Primary Aim

To evaluate whether standardized sulforaphane-rich microgreen supplementation, delivered over 12 months to genetically at-risk young Keralites (20-40 years), produces measurable epigenetic reprogramming (Shannon J' elevation, methylation restoration) accompanied by reduction in subclinical CAD biomarkers (FABP, oxidative stress, HRV parameters) and cardiovascular risk scores --- thereby extending CAD-free years by a projected 5-10 years.

Secondary Objectives

- Validate FABP quantified by NIR spectroscopy as a sensitive early-warning biomarker for subclinical myocardial metabolic stress in the 20-40 age group.
- Characterise Kerala-specific genomic risk architecture across 14 districts using ACE, PCSK9, HMGCoA reductase, MTHFR, TCF7L2, FTO, PPARG, and MC4R SNP panels.
- Demonstrate sulforaphane-medicated NRF2 pathway activation quantifiable through serum MDA reduction and SOD elevation.
- Establish CardioSense HRV phenotyping (SDNN, RMSSD, Shannon entropy, triangular index, LF/HF ratio) as a continuous autonomic risk biomarker in pre-symptomatic CAD.
- Build and validate the HRIDAYALAYA Genomics, HRV, FABP, multi-omics, and Shannon J' nutritional intelligence.
- Generate district-wise epigenomic maps of CAD vulnerability across Kerala for State health policy translation.

3. INCLUSION & EXCLUSION CRITERIA

3.1 Inclusion Criteria --- Risk Factor Group (n = 250)

- Age 20-40 years, either sex, Kerala domicile >=5 years
- Presence of >=1 established CAD risk factor:
    - Hypertension: BP >=130/80 mmHg on two readings OR on antihypertensive therapy
    - Dyslipidaemia: LDL >130 mg/dL, TG >150 mg/dL, or HDL <40 mg/dL
    - Prediabetes/Type 2 DM: FBG 100-125 mg/dL or HbA1c 5.7-6.4% (prediabetes), or diagnosed T2DM
    - Obesity: BMI >=25 kg/m^2 (Asian cut-off) or waist circumference >=90 cm (M) or >=80 cm (F)
    - Current smoker or <=2 years cessation
    - Strong family history: first-degree relative with CAD <55 years (M) or <65 years (F)
- Willing to consume standardized microgreens 20-50 g/day for 12 months
- Able to provide informed written consent
- Smartphone access for CardioSense HRV monitoring

3.2 Inclusion Criteria --- Control Group (n = 250)

- Age 20-40 years, matched for sex and district to risk factor participants
- No known hypertension, dyslipidaemia, diabetes, or obesity
- BMI <25 kg/m^2, BP <120/80 mmHg, FBG <100 mg/dL, HbA1c <5.7%
- Non-smoker, no family history of premature CAD
- Willing to undergo same assessment battery

3.3 Exclusion Criteria

- Established CAD: prior MI, PCI, CABG, or angiographic CAD
- Known cardiomyopathy, heart failure (LVEF <50%), or valvular heart disease
- Chronic kidney disease (eGFR <60 mL/min/1.73m^2)
- Liver disease (ALT/AST >2x ULN)
- Autoimmune, inflammatory, or malignant conditions
- Thyroid disorders (uncontrolled hypothyroidism or hyperthyroidism)
- Pregnancy, lactation, or planned pregnancy within 12 months
- Currently on antioxidant supplements (Vitamin C >500 mg/day, Vitamin E >400 IU/day, NAC)
- Known allergy to Brassica family vegetables

- Inability to comply with dietary protocol (swallowing disorder, GI malabsorption)
- Participation in any other clinical trial within 6 months
- Psychiatric illness precluding informed consent